

TECHNICAL REPORT

Structure First Proof Engineering for MIPLIB Open Instances

Technical report on the MIPLIB open problem repository

Evidence graded synthesis of 112 studied instances, 32 repository supported global conclusions, and the structure driven methods behind the strongest proofs.

Evidence date	18 September 2026 UTC (17 September 2026 PDT)
Published baseline	Commit 89a8abd dated 16 September 2026
Supplement	Local first five review dated 18 September 2026 UTC, not yet committed
Audience	Mixed integer optimization, exact algorithms, automated reasoning, and AI assisted mathematics

Prepared from the repository evidence archive. Official MIPLIB status was not live refreshed for this report.

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Executive summary

The durable result of the `MIPLIB_openproblem` project is a research protocol, not a single solver setting. The successful cases repeatedly replace a large anonymous mixed integer linear program with a smaller combinatorial object whose relationship to the original model can be checked. Parity, interval orders, spanning trees, symmetry quotients, dynamic programs, and proof producing SAT all appear in the archive. The final claim is then carried back to the untouched MPS through a feasible witness, a valid global bound, and an explicit evidence grade.

On the fixed 4 September 2026 snapshot of 217 officially open MIPLIB instances, the repository studied 112 current open instances and retained one historical calibration instance. It reports 32 global conclusions among the 112 studied instances: 30 optimal values and two infeasibility results. Eighty studied instances remain globally unresolved. These numbers describe repository evidence, not updates to the official MIPLIB status pages, and they do not estimate a success probability because instance selection and research budgets were nonuniform [1].

The repository also records 21 improvements over existing MIPLIB v36 primal values, four first finite incumbents, and 41 improved lower bounds relative to a frozen historical COPT 10 hour table. These are three different, overlapping comparisons. The primal and dual counts do not share a baseline, and the lower bound comparison is not a same hardware or same budget solver benchmark.

The central findings are:

1. Model size often overstates the dimension of the decisive combinatorial core.
2. Construction and proof are different pipelines. A strong feasible solution does not prove a lower bound, and a lower bound does not supply a witness.
3. Reduced model optimality, local infeasibility, and unchecked SAT unsatisfiability are scoped statements, not original model conclusions.
4. Proof objects and original model audits are part of the algorithm, not postprocessing.
5. Negative results are valuable only when their scope is preserved and they change the next experiment.
6. The structure first workflow is complementary to generic branch and bound. In a nonrandomized 20 instance comparison, its structure arm recorded better selected primals and two exact closures, while the comparison arm often retained stronger numerical dual bounds. Those two closures belong to the separate comparison archive, not the canonical 112 instance inventory.

1 Problem setting and scope

For a minimization problem, the project uses the basic proof obligation $\mathbf{LB} \leq \mathbf{OPT} \leq \mathbf{UB}$, with global optimality established only when a valid lower bound and a strictly feasible upper bound coincide.

The phrase **open instance** is frozen to the MIPLIB open list downloaded on 4 September 2026 [12]. It means the official record at that snapshot did not contain a global closure. It does not mean that no later local computation, external construction, or published theorem exists. The project therefore separates three statuses:

- the official status in the frozen MIPLIB snapshot;
- the conclusion supported by the repository evidence; and
- whether that conclusion has a portable, independently replayable proof.

The authoritative numerical baseline for this report is the 16 September 2026 presentation [4] and the 14 September dual bound audit [3], not the older coverage taxonomy still embedded in parts of the root README. The later audit upgrades the five `fhmw-schedule-pair` instances from solver based closures to exact branch and dual certificates and corrects several terminal decimal digits.

The uncommitted 18 September first five review is treated as a supplement [11]. It hardens the `assign1-10-4` and `fhmw-binschedule0` checks and fills a proof scope gap in `cvrpp-n16k8vrpi` by covering every nonempty route count from one through fifteen. Because that package is not yet in the published commit, this report marks the CVRPP classification as pending integration even though the supplemental audit again obtains value 450.

2 Evidence model

The project deliberately avoids one undifferentiated solved label. Its evidence classes answer different questions.

Evidence class	Opt.	Infeas.	Total	What is checked
Self-contained portable exact certificate	17	1	18*	Original model mapping, exact bound, witness, and solver-free replay
Checked exact result with large trace stored outside Git by hash	1	0	1	Compact mapping and logs are local; the 2.65 GiB proof trace must be obtained or regenerated
Specialized exact exhaustive computation without standalone trace	5	0	5	Exact program and output are archived, but not a small independent proof object
Exact model crosswalk plus published result	2	1	3	Local mapping is checked; the external theorem or computation is accepted rather than replayed
Floating-point solver closure	3	0	3	Strict primal audit plus commercial global lower bound; no portable exact lower bound trace
Mixed database and computational argument	2	0	2	Exact projection is combined with external benchmark values and commercial exclusions
Total	30	2	32	

Table 1: Reconciled working tree evidence classification for the 32 repository supported global conclusions.

The 18 portable exact conclusions include the five exact FHNW scheduling certificates completed in the 14 September audit. The table gives a reconciled working tree classification, not a commit only classification. The asterisk marks one material qualification: the committed CVRPP proof fixed eight routes even though the compiled model permits up to fifteen vehicle slots. The uncommitted supplement closes that scope gap; until it is integrated, the portable classification should carry this audit note. The three literature transfers rely on exact local crosswalks to externally published results [15][16][17].

This evidence model also defines what weaker artifacts do not prove:

- A feasible candidate accepted at a loose tolerance is not yet a strict incumbent.
- **OPTIMAL** on a projection proves the original problem only if the projection direction and lifting argument are correct.
- **INFEASIBLE** in a fixed neighborhood excludes only that neighborhood.

- A time limit is neither feasibility nor infeasibility evidence.
- SAT or pseudo Boolean UNKNOWN is not an unsatisfiability result.
- UNSAT is not portable without a checked encoding implication and a replayed proof trace.
- A paper value transfers only after the exact instance identity and model crosswalk are established.

3 Structure driven research protocol

The repository’s recurring protocol can be summarized as follows.

1. **Freeze the target claim.** Decide whether the goal is a primal improvement, a dual bound, feasibility, infeasibility, or global closure.
2. **Read the untouched MPS.** Record variable domains, objective support, row senses, coefficient patterns, sparsity, repeated columns, numerical scales, and candidate graph or block structure.
3. **Recover semantics and provenance.** Identify the underlying problem family, sibling formulations, public solutions, and relevant theorems without treating names or metadata as proof.
4. **Form falsifiable structural hypotheses.** Each proposed reduction or inequality is paired with a small counterexample search or exact consistency check.
5. **Run bounded pilots.** Experiments have explicit scope and stop conditions. A failed pilot updates the model of the problem rather than disappearing from the record.
6. **Separate the two pipelines.** The primal pipeline constructs and repairs witnesses. The dual pipeline develops relaxations, counting arguments, dynamic programs, proof trees, or cutoff refutations.
7. **Return to the original MPS.** Reconstruct every variable needed for a full solution, verify every row, bound, and integrality condition, and audit objective constants and signs.
8. **Grade the evidence.** State whether the result is exact, exhaustive, theorem dependent, solver numerical, local, or conditional.
9. **Stop or change representation.** Repeated null steps, exhausted neighborhoods, invalid cuts, or incomplete proof traces are signals to change the state space rather than lengthen the same run.

The key design principle is that expression complexity and combinatorial complexity differ. A model with hundreds of thousands of columns can be controlled by a few hundred selections, a path cover in an interval graph, a small separator state, a finite target cutoff, or a handful of coupling rows. The research task is to find the smallest representation that preserves the direction of inference needed for the claim.

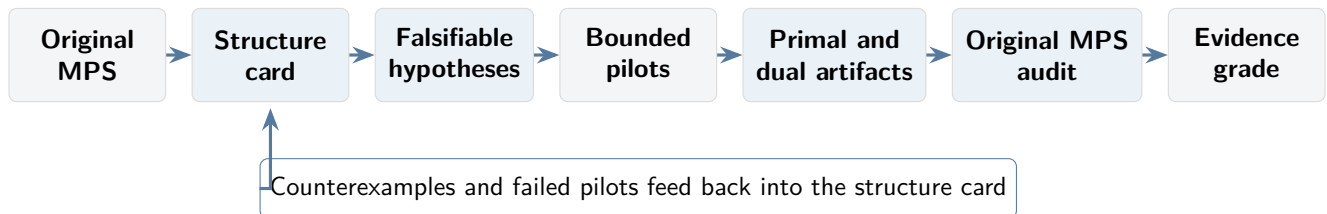


Figure 1: Structure first research workflow with explicit falsification and return to the original MPS.

4 Aggregate results

4.1 Coverage

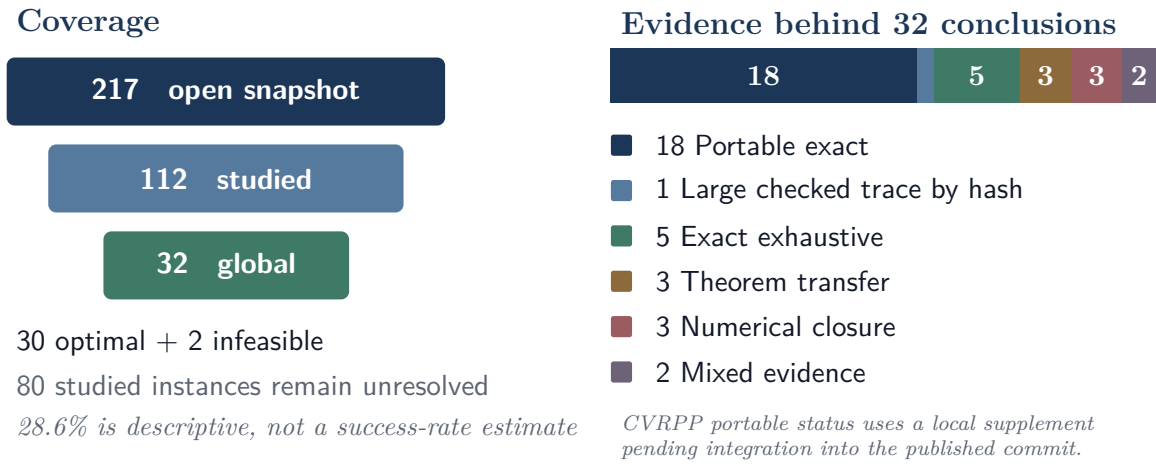


Figure 2: Frozen campaign coverage and evidence mix. Evidence classes preserve materially different dependencies.

Metric	Count	Interpretation
Open instances in frozen 4 September snapshot	217	Historical universe, not a live status query
Current open instances studied	112	Plus one historical calibration instance, <code>rmine14</code>
Repository supported global conclusions	32	30 optimal and 2 infeasible
Studied but globally unresolved	80	Includes useful primal, dual, structural, and negative results
Closure fraction within studied set	28.6 percent	Descriptive only; selection and budgets were nonuniform

Table 2: Frozen campaign coverage and interpretation.

4.2 Primal and dual progress

The repository counts 21 numerical improvements over the official MIPLIB v36 solution baseline [13] and four first finite incumbents, for 25 primal advances. The count includes two `fastxgemm` improvements derived from an attributed external construction; the independent project contribution is the truncation, mapping, and strict verification, not authorship of the parent construction. The published R23 construction is reproduced but not counted as a project primal improvement.

The 41 dual advantages compare repository bounds with displayed values in a frozen historical COPT 10 hour table [14]. They mix exact certificates, theorem transfers, commercial solver bounds, and numerical evidence. They therefore measure archived bound progress, not solver speed. Two notable exact but still nonclosing bounds are `graphdraw-mainerd` $\geq 31,882$ and `graphdraw-opmanager` $\geq 58,950$.

4.3 Verification performed for this report

Two solver free replay suites were executed in temporary directories during preparation of this report.

- The 14 September portable suite completed all 15 tasks with exit status zero. It replayed five FHNW scheduling certificates, `ramos3`, `sorrell17`, two `graphdraw` certificates, the exact large route CVRPA bound, negative controls, and both CVRP projection and primal checks. No commercial optimizer was called.
- The frozen ten-instance publication verified all 259 file hashes and completed all ten requested checks. It replayed compact combinatorial certificates, zero-tolerance witnesses, genus catalogues, an exact branch tree, and an LRAT proof. It launched no optimization search.

These replays do not cover the external theorem computations, the absent 2.65 GiB BPPC proof trace, specialized exhaustive computations lacking standalone traces, or historical commercial solver searches.

5 Four proof architectures

5.1 Parity and a matching witness in `allcolor58`

`allcolor58` has 84,376 variables and 197,154 rows, but its decisive lower bound is a parity argument. Every legal package has even total capacity, and assignment counts are integral, so each store receives an even total quantity. Exactly 42 stores have odd total demand. Every such store must incur at least one unit of overstock or understock cost, proving a global lower bound of 42.

The value 42 and a construction seed were already present in external or literature information. The repository contribution is therefore not a claim of first discovery: it reconstructs the parity lower bound, produces a replayable full witness from a compact search, and subjects that witness to exact original model audits [5].

The primal side is separate. A factorized search constructs fourteen package configurations whose allocations give exactly one unit of overstock at each odd demand store and no other deviation. Two independent exact audits verify all variables and all original rows with zero violation. Thus the parity lower bound and the strict witness meet at 42.

This case also records an important historical correction. Earlier objective zero candidates failed independent verification. They remain in the archive as failed experiments rather than being rewritten as progress. The eventual proof did not come from more aggressive tuning of those candidates; it came from changing the representation.

5.2 Spanning trees and exact subset dynamic programming in `dfn-bwin-DBE`

The network design model has ten nodes and positive demand for every ordered pair. Any feasible installed support must therefore be connected. A connected support with nine physical edges is a spanning tree. Any support with ten or more edges costs at least the sum of the ten cheapest positive opening costs, 81,222.40, already greater than the candidate value 73,623.79.

It remains to optimize over spanning trees. Removing a tree edge creates a cut; unique tree paths force all cross cut demand through that edge. The two directional totals determine the minimum feasible capacity option and its exact cost. Two independent subset dynamic programs enumerate the rooted tree decompositions and obtain 7,362,379 cents for every root. The generated tree solution passes an exact original model audit, completing the proof of 73,623.79 [6].

The conceptual reduction is global, not heuristic. The ten-edge cost bound excludes every denser support, while the dynamic program covers every tree support. The proof is not a claim that a good solution happened to be a tree.

5.3 Interval orders and load parity in `fhnw-binschedule2`

The original MPS is a six machine makespan formulation with 497 items, 114 ordered positions, 1,614 local OR variables, 1,908 selector windows, and 228 difference indicators. Seventy six items can use only the two 39 position endpoint machines.

Recovering the selector windows exposes an interval order. A set of 35 intervals all contains the same interior point, so its members are pairwise incomparable. If endpoint machine k receives n_k mandatory items and activates D_k differences, its sequence can cover at most $40 - n_k + D_k$ chains. Summing over both endpoints forces at least 31 active differences.

The mandatory item costs total 4,546 and every difference costs 10, so the two endpoint loads sum to at least 4,856. Endpoint loads lie on an even integer lattice. If the makespan were below 2,428, both endpoint loads would be at most 2,426 and their sum at most 4,852, a contradiction. The public solution attains 2,428 and passes exact checking of all 6,503 rows [7].

The proof combines two structures that a generic relaxation largely obscures: an interval antichain controls the number of breaks, and a lattice restriction removes the remaining numerical slack.

5.4 Tight profiles and proof-producing SAT in `bppc6-06`

`bppc6-06` chooses one start position for each of twenty items on a horizon of 300 and minimizes the maximum profile height. A known placement has height 208. The lower bound task is therefore the finite decision question: can height 207 occur?

An exact subset enumeration finds nonnegative integer weights for the items such that every subset with physical height at most 207 has weight at most four. The total weighted length is exactly four times the horizon, so every positive width profile segment in a hypothetical height 207 placement must have weight exactly four. This creates a much stronger Boolean encoding than the raw height constraints alone.

The resulting CNF has 414,391 Boolean variables and 1,541,322 clauses. CaDiCaL reports unsatisfiable, `drat-trim` verifies the proof, and a separate checker replays 1,001,496 LRAT lemmas through the final empty clause. Two earlier incomplete proof logs are explicitly rejected. The exact 208 placement passes the original MPS audit, so the checked cutoff refutation and witness establish the optimum [8].

The limitation is equally explicit: the complete CNF, DRAT, and LRAT artifacts total about 2.65 GiB and are stored outside Git by byte size and hash. The repository contains the encoder, compact mapping, manifests, and replay summaries, but an independent reviewer must obtain or regenerate the large files for end-to-end replay.

6 Proof correction and negative evidence

The archive is unusually useful because it records incorrect turns and their scope.

An early FHNW scheduling route interpreted a latest start bound as a completion deadline. That missing duration term produced invalid energy cuts and apparent zero gaps. Original model counterexamples forced the dual claims to be withdrawn. The final five proofs use safe release suffix relaxations, complete binary branch trees, exact rational leaf duals, and full original MPS witnesses. The correction changed both the mathematics and the evidence grade.

Other recurring failure modes include:

- parsers that miss indented RHS records;
- row orientation errors in pseudo Boolean encodings;
- reductions that are valid as relaxations but are mistakenly treated as liftable equivalences;
- time-limited SAT or MILP runs written as if they had excluded a target;
- a fixed support or local Hamming ball confused with the full feasible region;
- floating-point solutions whose apparent improvement vanishes after exact row and integrality checks;
- proof generation logs truncated before the final certificate is flushed.

The response is a persistent experimental ledger. Each result states the precise subproblem, target, time limit, terminal status, mapping direction, and verification performed. A failed experiment is valuable when it rules out a mechanism or identifies the minimum scale of the next move. It is not evidence of global impossibility.

7 Observational workflow comparison

A separate historical study compared a structure research skill with a run that did not invoke it on twenty instances [9]. Under its absolute $1e-7$ selected primal rule, the structure arm recorded better values on eleven instances and tied on nine, and it recorded two exact global closures. Six selected values were tolerance indexed rather than strict incumbents; against strict public baselines, the improvement counts were four for the structure arm and two for the comparison arm. The comparison arm retained stronger dual bounds on twelve of twenty instances and smaller common gaps on eleven of twenty. Its two reported closures, `ns1456591` and `neos-3682128-sandon`, remain in that separate comparison archive and are not integrated into the canonical 112 instance, 32 conclusion inventory.

This is observational evidence, not a causal estimate. The runs were not randomized, blinded, or equal budget; machines, starts, solver states, and stopping decisions differed. The defensible conclusion is operational:

- structure first work is valuable for semantic recovery, new representations, primal construction, certificate design, and original model audit;
- a protected generic solver budget remains valuable for numerical dual progress when the structural route does not close the instance.

The comparison therefore supports a hybrid research design rather than replacing branch and bound with a language model.

8 Remaining research frontier

The 80 unresolved studied instances contain more than unfinished solver runs. Many already have exact intervals, verified reductions, completed local exclusions, or proof kernels. The continuation ranking [10] argues that the best next target is not necessarily the smallest displayed gap. Priority should instead go to an instance already reduced to a small auditable decision problem with a plausible certificate route.

The leading continuation portfolio in the 9 September snapshot is:

1. `graph40-20-1rand`, where a sibling SAT and LRAT template suggests a direct 16 cut refutation after graph identity is certified;

2. `genus-sym-g62-2`, where only a small recorded frontier remains in an exact face search;
3. `genus-g61-25`, where the next even face target is finite and checkpointed;
4. `ns1828997`, where a very small strict gap and objective lattice make a cutoff proof plausible after exact reproduction.

The ranking is a research allocation heuristic, not a calibrated probability of solution. Each continuation needs a stage gate that confirms the required inputs, mapping, and proof backend before a long run.

9 Threats to validity

9.1 Selection and budget bias

The 112 studied instances were not sampled randomly, and the project preferentially pursued visible structure, tractable certificates, and promising siblings. Research budgets varied. The ratio 32 divided by 112 is therefore a historical closure fraction, not a general success rate.

9.2 Evidence heterogeneity

Exact certificates, specialized exhaustive programs, published theorems, commercial zero gap logs, and mixed database arguments do not provide the same portability. Aggregate counts must preserve their evidence mix.

9.3 Official status lag

Repository conclusions may postdate the frozen MIPLIB solution file or remain unreviewed upstream. This report never treats a repository closure as an official status change.

9.4 External and missing artifacts

The large BPPC trace is external by hash; literature computations were not replayed; five exact exhaustive results lack compact independent traces; three closures are floating-point solver closures; two mixed CVRP conclusions additionally depend on commercial route-count exclusions; and seventeen instance folders retain provenance rather than a local MPS copy.

9.5 Reproducibility details

Some scripts require Python 3.10 or later. The shared exact solution checker prints violation counts but does not always fail through its process status, so reviewers must inspect the recorded counts. The `minutedispatchstrategy` archive retains a conflict with an old unavailable PDF primal value, and the `cmflsp40-36-2-10` numerical result lacks its original global runner.

9.6 Source synchronization

The root README contains stale evidence counts and stale FHNW labels. The 14 September audit and 16 September presentation are the later authorities for evidence classes and exact decimals. The report preserves this precedence instead of averaging conflicting summaries.

10 Reproducibility contract

A publication quality result should archive the following components.

1. The exact original MPS or a stable official URL, byte count, and cryptographic hash.
2. The objective sense, objective offset, variable domains, row senses, and numerical scaling.
3. A proof of the direction from every original feasible point to the reduced object used for the bound.
4. A lifting or reconstruction procedure when reduced feasible points are used to build original witnesses.
5. A complete original model solution file with exact or explicitly bounded tolerance checks of every row, bound, and integrality condition.
6. A certificate whose semantics are independent of the search program whenever practical.
7. An independent checker, negative controls, software version, and replay command.
8. Hashes for all frozen source, certificate, and checker files.
9. An evidence label and an explicit list of what was not replayed.
10. A distinction between the repository result and the official benchmark status.

The archive already follows much of this contract, but the supplemental audits show why it matters. A correct value can coexist with a checker scope gap. Independent parsing and a second proof route convert that value into a more trustworthy result.

11 Conclusion

The `MIPLIB_openproblem` repository does not support the claim that a language model is a universal MILP solver. It supports a narrower and more useful thesis: AI assisted research can contribute when it helps translate an opaque formulation into a smaller, falsifiable, and auditable mathematical object, while exact scripts and solvers enforce the boundary between a plausible idea and a proved statement.

The strongest examples are not parameter searches. They are changes of state space: even capacity becomes parity, network design becomes a tree dynamic program, scheduling becomes an interval chain cover, and a narrow packing cutoff becomes a proof producing SAT instance. The reusable scientific asset is the loop that connects those representations back to the original MPS and assigns each conclusion the evidence grade it actually earns.

A Repository reported global conclusions

Abbreviations: PE portable exact certificate; HP checked exact proof with large trace external by hash; EX specialized exact exhaustive computation; LT exact crosswalk plus published theorem or result; NS numerical solver closure; MX mixed database and computational evidence.

Table 3: Repository reported global conclusions and evidence grades.

Instance	Conclusion	Grade
assign1-10-4	OPT = 422	PE
cvrpp-n16k8vrpi	OPT = 450	PE*
dfn-bwin-DBE	OPT = 73,623.79	PE
fhnw-binschedule0	OPT = 15,958	PE
fhnw-binschedule1	OPT = 55,158	PE
fhnw-binschedule2	OPT = 2,428	PE
fhnw-schedule-paira100	OPT = -15.113116512593401992	PE
fhnw-schedule-paira200	OPT = -19.369961204645205810	PE
fhnw-schedule-pairb200	OPT = -19.369961204645205810	PE
fhnw-schedule-paira400	OPT = -36.276966742368865136	PE
fhnw-schedule-pairb400	OPT = -36.276966742368865136	PE
neos-3009394-lami	OPT = 5.5	PE
neos-2629914-sudost	OPT = 48,180	PE
allcolor58	OPT = 42	PE
graph40-80-1rand	OPT = -7	PE
graph40-40-1rand	OPT = -9	PE
neos-4355351-swalm	OPT = 33.45763	PE
bppc6-06	OPT = 208	HP
neos-4360552-sangro	OPT = -8	EX
ns1631475	OPT = 11,100	EX
genus-sym-g31-8	OPT = -23	EX
genus-g31-8	OPT = -23	EX
genus-sym-grafo5708-48	OPT = -21	EX
circ10-3	OPT = 242	LT
a2864-99blp	OPT = -257	LT
neos-3594536-henty	OPT approximately 401,092	NS
neos-2978205-isar	OPT approximately -11.925808687	NS
minutedispatchstrategy	repository closure approximately 3,109.9034777596576	NS [†]
cvrpb-n45k5vrpi	OPT = 751	MX
cvrpa-n64k9vrpi	OPT = 1,401	MX
fhnw-binpack4-58	infeasible	PE
pythagor7825	infeasible	LT

**CVRPP's portable grade uses the 18 September local scope supplement, which is pending integration into the published commit. [†]The minutedispatchstrategy closure retains an unresolved discrepancy with an older unavailable-PDF primal value.*

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17. Published Boolean Pythagorean triples result used by the `pythago7825` crosswalk